Ratios, Rates & Data

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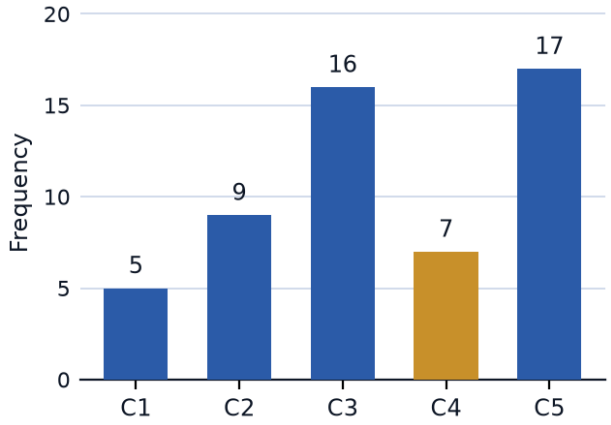
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Ratios and proportions

Simplify the ratio $18 : 24$ to lowest terms.

2 [Foundational] • Target: Ratios and proportions

Solve the proportion $\frac{5}{8} = \frac{x}{40}$.



3 [Foundational] • Target: Ratios and proportions

A recipe requires flour and sugar in the ratio $3 : 2$. If 12 cups of flour are used, how much sugar is needed?

4 [Foundational] • Target: Unit rates

A car uses 8 gallons of gas to travel 224 miles. Find its fuel efficiency in miles per gallon.

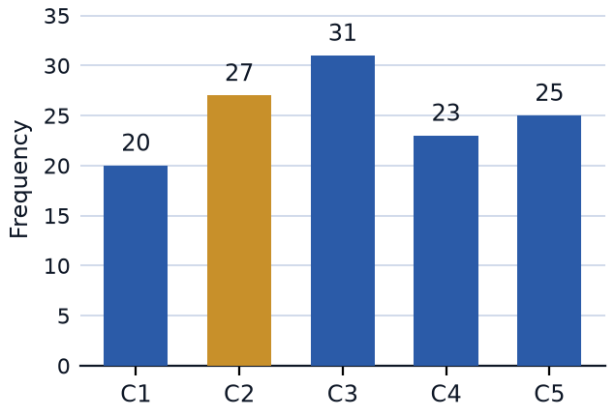
5 [Foundational] • Target: Unit rates

A printer produces 150 pages in 6 minutes. Find its rate in pages per minute, and use it to find how long it takes to print 400 pages.

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6 [Foundational] • Target: Percentage change and applications

A price increases from 60 to 75 dollars. Find the percentage increase.



7 [Foundational] • Target: Percentage change and applications

A jacket originally 120 dollars is discounted 30 percent. Find the sale price.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Percentage change and applications

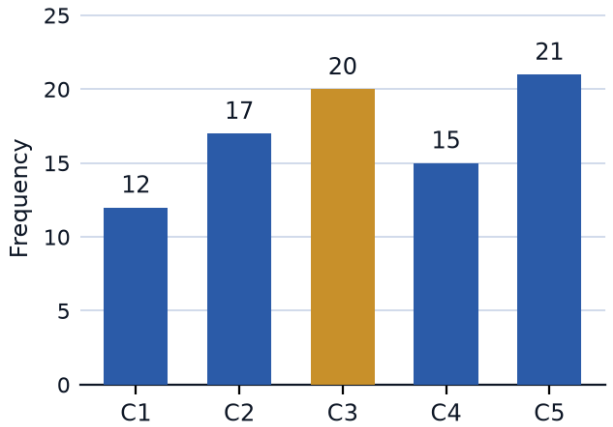
A population decreases by 15 percent one year, then increases by 15 percent the next. Is the final population the same as the original?

9 [Intermediate] • Target: Percentage change and applications

An investment of 2000 dollars grows to 2300 dollars after one year. Find the percentage increase.

10 [Intermediate] • Target: Weighted averages

A student scores 80 on a test worth 40 percent of the grade and 90 on a test worth 60 percent of the grade. Find the weighted average grade.



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11 [Intermediate] • Target: Weighted averages

A store sells 3 items at 10 dollars each and 7 items at 20 dollars each. Find the average price per item sold.

12 [Intermediate] • Target: Scale and similar figures

A model car is built at a scale of $1:24$. If the actual car is 4.8 meters (480 cm) long, find the length of the model, in centimeters.

13 [Intermediate] • Target: Scale and similar figures

A map has a scale of 1 cm to 5 km. If two cities are 8.5 cm apart on the map, find the actual distance between them.

14 [Intermediate] • Target: Interpreting data tables

A table shows the hours worked by 4 employees in a week: A: 32, B: 40, C: 28, D: 36. Find the total hours worked and the mean hours per employee.

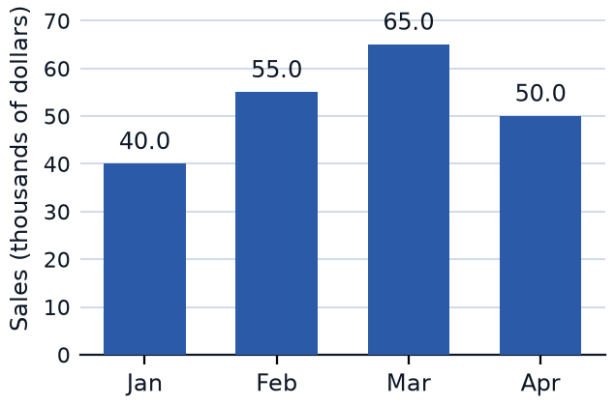
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Interpreting data tables • Cross-Lesson: 5.01 Statistics & Probability

A table shows quarterly revenue, in thousands of dollars: Q1: 50, Q2: 65, Q3: 70, Q4: 55. Find the percentage of annual revenue earned in Q3, to the nearest tenth of a percent.

16 [Challenging] • Target: Bar graphs and data displays • Cross-Lesson: 6.01 Geometry

The bar graph shows a company's monthly sales, in thousands of dollars. Find the percentage increase in sales from January to March.



17 [Challenging] • Target: Proportional reasoning in context • Cross-Lesson: 7.01 Trigonometry & Additional Topics

A factory produces 450 units using 6 machines running for 8 hours. Assuming the same rate per machine-hour, how many units would 9 machines produce in 8 hours?

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18 [Challenging] • Target: Proportional reasoning in context • Cross-Lesson: 8.01 Exponentials & Logarithms

If 5 workers can build a fence in 12 days, how many days would it take 4 workers, assuming the same work rate per worker?

19 [Challenging] • Target: A well-designed word problem: customer growth for a small business • Cross-Lesson: 1.01 Heart of Algebra

This question has three parts. In the first quarter, a business had 240 customers, and 180 of them made a repeat purchase. (a) Find the repeat-purchase rate as a percentage. (b) In the second quarter, the number of customers grew by 25 percent, and the repeat-purchase rate held steady at the percentage found in part (a). Find the number of repeat customers in the second quarter. (c) The business sets a goal to raise the repeat-purchase rate to 80 percent by the third quarter, while keeping the same total number of customers as the second quarter. Find how many additional repeat customers (beyond part (b)'s total) are needed to reach this goal.

20 [Challenging] • Target: Unit rates • Cross-Lesson: 2.01 Quadratics & Polynomials

In a course grading system, Homework is weighted 20%, Midterm Exam is weighted 35%, and Final Exam is weighted 45%. A student scored 92 on Homework and 76 on the Midterm Exam. What is the minimum score the student must achieve on the Final Exam to earn an overall course average of at least 84?

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Ratios and proportions

A budget decreases from 40 to 20. Find the percentage decrease.

a. 50%

b. 55%

c. 100%

d. 45%

22 [Foundational] • Target: Unit rates

A population increases from 220 to 264. Find the percentage increase.

a. 40%

b. 20%

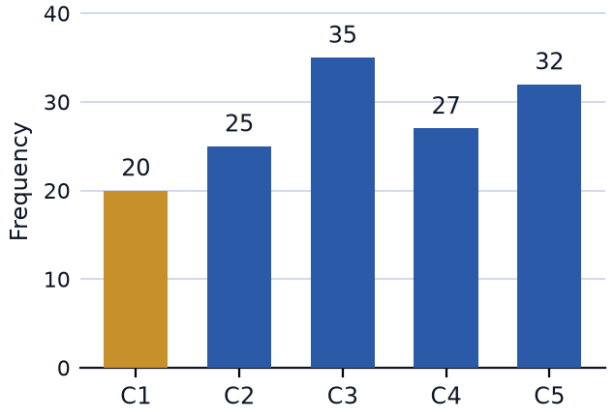
c. 25%

d. 15%

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23 [Foundational] • Target: Percentage change and applications

An attendance figure increases from 240 to 312. Find the percentage increase.



a. 30%

b. 25%

c. 35%

d. 60%

24 [Foundational] • Target: Weighted averages

A subscription count increases from 20 to 29. Find the percentage increase.

a. 40%

b. 45%

c. 90%

d. 50%

25 [Foundational] • Target: Scale and similar figures

A membership total increases from 60 to 78. Find the percentage increase.

a. 30%

b. 35%

c. 25%

d. 60%

26 [Foundational] • Target: Interpreting data tables

A membership total decreases from 60 to 30. Find the percentage decrease.

a. 55%

b. 100%

c. 45%

d. 50%

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27 [Foundational] • Target: Bar graphs and data displays

A population decreases from 260 to 169. Find the percentage decrease.

a. 30%

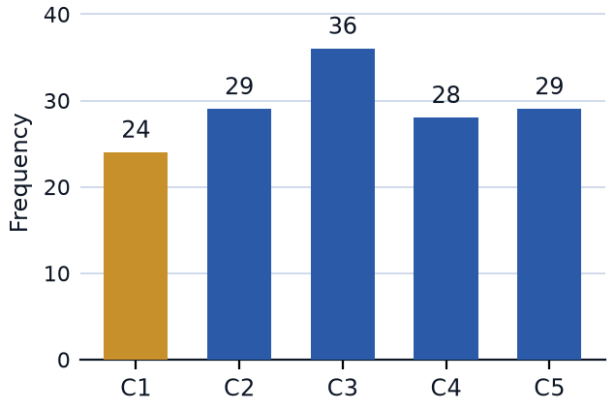
b. 70%

c. 35%

d. 40%

28 [Foundational] • Target: Proportional reasoning in context

A price increases from 260 to 377. Find the percentage increase.



a. 40%

b. 90%

c. 45%

d. 50%

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: A well-designed word problem: customer growth for a small business

An attendance figure decreases from 300 to 180. Find the percentage decrease.

a. 80%

b. 45%

c. 35%

d. 40%

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30 [Intermediate] • Target: Ratios and proportions

A student scores 60 on an assignment worth 75% of the grade and 100 on an assignment worth 25% of the grade. Find the weighted average grade.

a. 80

b. 70

c. 67

d. 73

31 [Intermediate] • Target: Unit rates

If 9 workers can complete a task in 24 days, how many days would it take 8 workers, assuming the same work rate per worker?

a. 29

b. 27

c. 24

d. 25

32 [Intermediate] • Target: Percentage change and applications

The bar chart shows a small business's quarterly income (in 1000s) for one year. In which quarter did they earn the smallest income?

a. 1st quarter

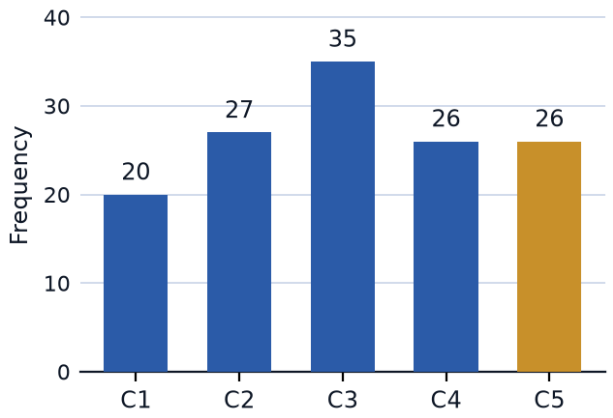
b. 2nd quarter

c. 3rd quarter

d. 4th quarter

33 [Intermediate] • Target: Weighted averages

The population of a town increased by 20% in the first year and then decreased by 15% in the second year. What was the net percentage change over the two years?



a. Increase of 2%

b. Increase of 5%

c. Decrease of 2%

d. No change

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34 **[Intermediate]** • **Target: Scale and similar figures**

The price of gasoline drops by 15% and then rises by 20%. What is the net overall percentage change?

a. Increase of 2%

b. Increase of 5%

c. Decrease of 2%

d. Increase of 1%

35 **[Intermediate]** • **Target: Interpreting data tables**

A survey found that 34% of respondents trust TV News most, and 22% trust Radio most. If a similar survey was conducted on 120 people with the same percentage breakdown, approximately how many people would say radio or TV news is most trustworthy?

a. 54

b. 61

c. 67

d. 72

36 **[Intermediate]** • **Target: Bar graphs and data displays**

The bar chart shows the percentage of primary students who like each subject. Which subject had the lowest percentage of likers?

a. Math

b. English

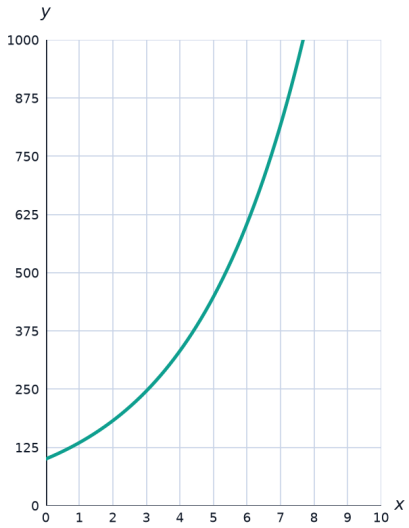
c. Science

d. History

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37 **[Intermediate]** • **Target: Proportional reasoning in context**

By what percentage should we raise the price of a good to return to its initial price knowing that it was subject to a 20% discount 2 weeks ago?



a. 15%

b. 20%

c. 25%

d. 30%

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 **[Challenging]** • **Target: A well-designed word problem: customer growth for a small business** • **Cross-Lesson: 8.01 Exponentials & Logarithms**

The bar chart below shows the number of students who participated in different sports. The number of students who participated in swimming is unknown and represented by x. If the total number of students who participated in all sports is 120, and the number of students who participated in basketball, volleyball, and soccer are 40, 30, and 20 respectively, what is the value of x?

a. 20

b. 30

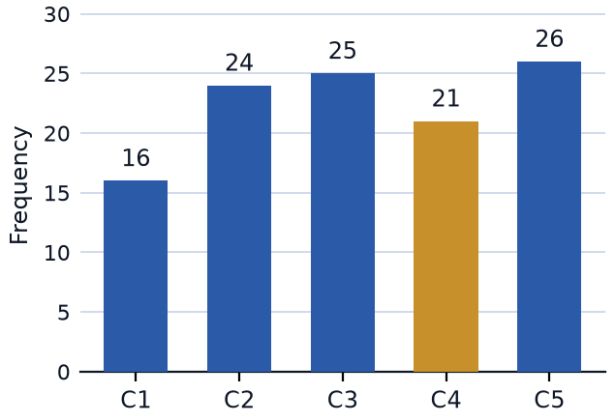
c. 40

d. 50

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39 [Challenging] • Target: Ratios and proportions • Cross-Lesson: 1.01 Heart of Algebra

The bar chart below shows the number of students who participated in different sports at a school. What is the ratio of the number of students who play basketball to the number of students who play volleyball?



a. 1:2

b. 2:3

c. 3:2

d. 3:4

40 [Challenging] • Target: Unit rates • Cross-Lesson: 2.01 Quadratics & Polynomials

A chemist mixes 30 mL of a 20% saline solution with 20 mL of a 40% saline solution. What is the saline concentration percentage of the resulting mixture?

a. 28%

b. 25%

c. 30%

d. 36%

41 [Challenging] • Target: Percentage change and applications • Cross-Lesson: 3.01 Functions & Graphs

According to the data published by “Worldometer”, the total cases of people infected by Covid-19 in Egypt on July 24, 2020, was 91,072, while in August 3, 2020 it went up to 94,640 cases. If we consider the graph of the change in number of total cases over the change in days is linear, what will be the total cases reported in August 9, 2020? (Note that there are 31 days in July).

a. 96,781 cases

b. 97,138 cases

c. 98,208 cases

d. 101,776 cases

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42 [Challenging] • Target: Weighted averages • Cross-Lesson: 5.01 Statistics & Probability

A survey of 300 parents asked which news source they trust most: TV News 102, Newspapers 45, Social Media 87, Radio 66. What percentage of parents trust social media most?

a. 15%

b. 22%

c. 29%

d. 34%

43 [Challenging] • Target: Scale and similar figures • Cross-Lesson: 6.01 Geometry

On a math test of 30 questions, Mona got 75% of the 12 geometry questions correct, 60% of the 10 algebra questions correct, and 25% of the 8 trigonometry questions wrong. What percentage of all the questions did Mona get correct?

a. 56.6%

b. 160%

c. 70%

d. 210%

44 [Challenging] • Target: Interpreting data tables • Cross-Lesson: 7.01 Trigonometry & Additional Topics

The table shows a city's population for two years. Which statement correctly describes the percent change from 2018 to 2020?

a. Decreased by 15%

b. Increased by 15%

c. Decreased by 10%

d. Increased by 10%

45 [Challenging] • Target: Bar graphs and data displays • Cross-Lesson: 8.01 Exponentials & Logarithms

A scale model of a building is built at a scale of $1:200$. If the model is 65 cm tall, what is the actual height of the building in meters?

a. 130

b. 13

c. 1300

d. 65

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